

IFC Power Control via RIBBO

Progress Report — June 2026

Myoungjin Oh

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Problem

Lee et al., TCOM 2025

- IFC power control, $\mathbf{x} \in [0, 1]^D$, Rayleigh fading
- **SE** (spectral efficiency, sum-rate): optimum at *box corners* — strong links full-on, weak links off
- **EE** (energy efficiency, rate / power): optimum in the *interior* — mid power

Method

- RIBBO (Reinforced In-Context BBO): a transformer that learns *how to optimize* from past attempts
- Key parts: regret-to-go (RTG) target token, hindsight regret relabeling (HRR)

Reference solution (defines the gap)

- SE: WMMSE (classical solver; *local-optimal*)
- EE: $D=3$ exhaustive grid; $D=10/20$ multi-start gradient (40 restarts; near-global)
- **Optimality gap (%)** = $100 \times \frac{\text{ref} - \text{achieved}}{\text{ref}}$; **lower is better** (0 = reference). As in Lee et al.

Evaluation

- Unseen channels = new Rayleigh $\text{CN}(0, 1)$ realizations (same distribution)
- 300 steps; behavior-algorithm (BA) pool = 3BA (Random, Hill-Climbing, CMA-ES) unless noted

Which BA pool works best depends on the dimension

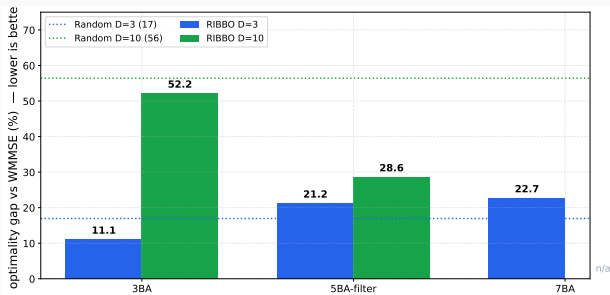
optimality gap (%) ↓	$D=3$	$D=10$
3BA	11.1	52.2
5BA-filter	21.2	28.6
7BA	22.7	—
Random	17.0	56.4

3BA = {Random, HC, CMA-ES}

5BA-filter = {HC, RegEvol, Eagle, CMA-ES, BoTorch}

7BA = 5BA-filter + Random + ShuffledGridSearch

- $D=3$: the simple 3BA pool is *best*; adding more algorithms hurts
- $D=10$: the curated 5BA-filter pool reduces the gap by 23.6 pts (52.2 → 28.6)
- Consistent with the RIBBO paper (use *well-performing* and *diverse* BAs), but the effect is dimension-dependent



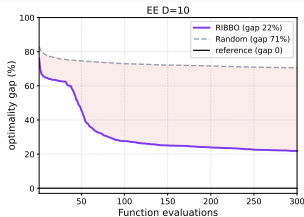
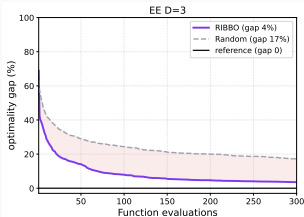
Bars: RIBBO gap to WMMSE. Dotted: Random. Lower = better.

EE's optimum is in the *interior* (mid power), so uniform sampling rarely finds it

gap (%) ↓	RIBBO	Random	Δ
EE $D=3$	3.5	17.2	13.7
EE $D=10$	21.9	70.6	48.7
SE $D=10$	52.2	56.4	4.2

Same 3BA pool; only the reward changes. Separate model per reward and dimension. Δ = Random – RIBBO.

- On EE $D=10$, Random's gap stays at 71% (uniform sampling rarely reaches the interior)
- RIBBO reduces it to 22% — a 48.7 pt smaller gap
- The clearest case for RIBBO in wireless



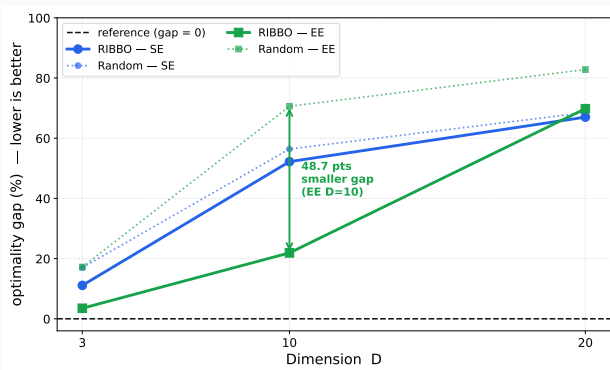
Shaded = how much smaller RIBBO's gap is; at $D=10$ Random's gap stays $\sim 71\%$.

Gap vs. dimension (SE & EE, 3BA)

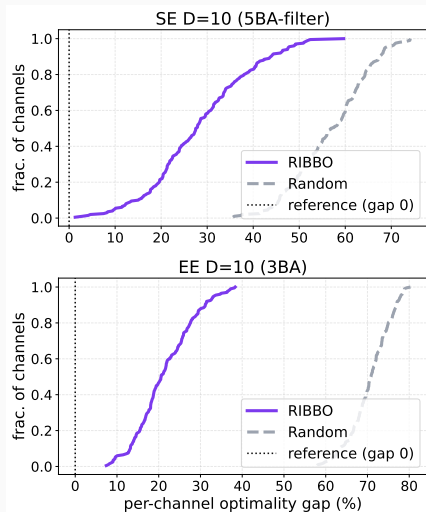
- Gap grows with D for both; one model *per dimension* (not dimension-agnostic)
- RIBBO's EE $D=10$ gap $\ll$ Random (48.7 pts); SE difference small

Reducing the gap (SE):

- Better algorithm choice: $D=20$
67.0 $\rightarrow$ 58.1 (−8.9)
- More steps: $D=10$ 27.1 $\rightarrow$ 19.0 (−8.1)
- But not for EE $D=20$ (69.9 $\rightarrow$ 69.7)
— needs more data



- Random search (= Lee et al.'s *Brute-force* baseline) plateaus above a 50% gap and rarely reaches the EE interior optimum; **RIBBO goes where uniform sampling cannot**
- Smaller gap than Random on **100%** of channels at $D=10$ (SE & EE), 99.5% at $D=20$
- Statistically stable: 95% CI $\pm < 1.5\%$ ($n=200-500$)
- EE advantage persists at $D=20$ (13 pt smaller, every channel)
- A local search reaches a ~ 0 gap from *any* start — with the channel model / gradients the problem is easy, so RIBBO's value is the *no-model* setting

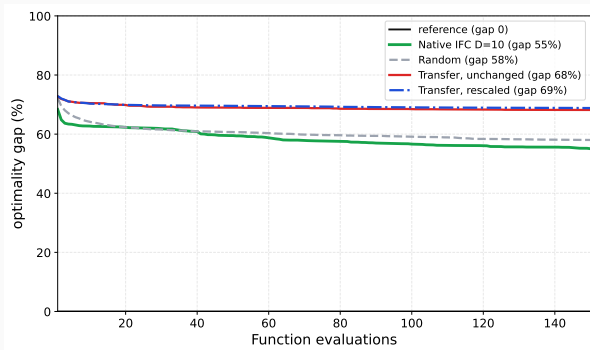


Use a Rastrigin-trained model on IFC

$D=10$ SE (no retraining)

Model	gap (%) ↓
Rastrigin model, unchanged	68.2
Rastrigin model, rescaled	68.8
IFC model (reference)	55.1
Random	~57

- Rescaling the target does *not* help (68.2 → 68.8)
- Our earlier guess (a scaling mismatch) was **wrong**
- Weak SE 3BA config here, so IFC-vs-Random is small — the point is *in-domain* vs. *transfer*
- ⇒ **in-domain IFC training is needed**



Rastrigin-trained (red/blue) have larger gaps than Random (gray); only IFC-trained (green) shrinks it.

- **The effect depends on the problem and its size.** For SE $D=3$ (on/off) simple methods already do well; for EE RIBBO shows a clearly smaller gap (48.7 pts at $D=10$).
- **Classical solvers need the channel model; RIBBO uses only scores.** WMMSE / fractional programming (FP) reach their reference solution with the model, and in our hybrid test a local search also closed the gap — so RIBBO's clearest role is the **no-model** setting.
- **Which training algorithms help is problem-dependent.** Dropping weak ones reduced the SE gap at high D (-23.6 pts) but increased the EE gap.
- **With a fixed setup, the gap grew with D ;** more data / steps and a better algorithm choice reduced it (SE).
- **Cross-problem transfer did not work:** a Rastrigin-trained model did not improve on IFC; in-domain training was needed.